$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}'_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big] - \mathrm{i}\gamma_0\big[\hat{X}, \big\{\hat{P}, \hat{\rho}_{\mathcal{S}}(t)\big\}\big] - 2M\gamma_0 k_{\mathrm{B}}T\big[\hat{X}, \big[\hat{X}, \hat{\rho}_{\mathcal{S}}(t)\big]\big], \tag{5.57}$$

where the frequency-shifted Hamiltonian $\hat{H}'_{\mathcal{S}}$ of the system is

$$\hat{H}'_{\mathcal{S}} = \hat{H}_{\mathcal{S}} + \frac{1}{2}M\widetilde{\Omega}^2\hat{X}^2 = \frac{1}{2M}\hat{P}^2 + \frac{1}{2}M\left[\Omega^2 - 2\gamma_0\Lambda\right]\hat{X}^2. \tag{5.58}$$